

Question Paper

Exam Date & Time: 03-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Computer Vision [IT_4408]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Modern industries and intelligent systems increasingly rely on computer vision for automation and decision-making. (5)
- A) Discuss one real-world application (for example: autonomous driving, medical diagnosis, or industrial inspection) and Discuss how computer vision transforms raw images into meaningful actions or insights
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Computer Vision plays a vital role in sports analytics - from tracking player movement to detecting ball trajectories. Analyse how Computer Vision enables such applications with a step-by-step approach (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Illustrate how edge detection techniques help in obtaining important structural details from digital images. Compare the working principles of the Laplacian and Canny edge detection methods (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Apply bit slicing, on the 4*4 matrix of your choice. Discuss the concept of bit slicing in detail. Write each pixel in 4-bit binary and give the bit-plane b3 (MSB) and bit-plane b0 (LSB) as binary matrices (0/1). (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) A self-driving car uses multiple cameras mounted at different angles. Apply your understanding of different camera models to explain how various camera models account for the differences in the field of view and distortion. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Analyse translation, rotation, and scaling operations to register two medical images of the same patient taken at different time intervals. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyze the Lucas-Kanade optical flow (point-based) to compute vehicle lane changes from dashcam video. Formulate how tracked points on lane markings are used to infer lateral vehicle motion and determine lane-change events. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Apply the mean-shift algorithm for real-time tracking of colored buoys in a marine robotics application. Explain how you would choose the kernel bandwidth, color space, and target model to be robust to lighting changes and reflections. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Analyze face recognition algorithms to design a secure office entry system that identifies authorized personnel. Explain how training data and distance thresholds affect system reliability. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Apply the BoW technique to categorize sports scenes (e.g., soccer, swimming, basketball) from video frames. Discuss how you would combine spatial pyramids or temporal information to enhance recognition accuracy (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4408:Computer Vision

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4408:Computer Vision

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)